Harshita Sharma

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Coding Profiles: Codeforces – harshitas LeetCode – starbus

Skills

Languages: Python, C, C++
Robotics: ROS2, Gazebo, PX4, , Blender, Pixhawk
ML: NumPy, Pandas, scikit-learn
OS/Tools: Ubuntu, Windows, Git, Raspberry Pi

Education

2024 - 2028	BTech, **IIT Bhilai, Mechanical Engineering**	**(Best SG: 8.6)** (CGPA: 8.39)
2023 - 2024	Class 12th CBSE	90.6%
2021 - 2022	Class 10th CBSE	90.2%

Work Experience and Projects

Autonomous Precision Landing of Quadrotor on Aruco Marker View Repo

- Developed and tested autonomous flight control algorithms for vision-based precision landing using ArUco marker detection
- Designed and assembled quadrotor hardware with Pixhawk 2.4.8 flight controller and Raspberry Pi 5 companion computer
- Created comprehensive simulation environment using PX4-Autopilot, Gazebo Classic, ROS2 Humble, and MAVROS for algorithm validation before hardware deployment

Tech stack- PX4-autopilot, MAVROS, Qgroundcontrol, Pixhawk 2.4.8, Raspberry pi5, Depth camera

Renewable Energy Forecasting Notebook

- Performed **data preprocessing** and **feature engineering** on time-series renewable energy dataset.
- Conducted comparative analysis of **solar vs wind energy generation**, identifying seasonal trends and relationships with **temperature and wind speed**.
- Visualized trends using **Matplotlib and Seaborn**, including temporal plots and correlation analysis.
- Provided insights useful for **renewable energy planning and power plant optimization**, addressing variability in energy supply.

Tech stack- Numpy, Pandas, scikit-learn, matplotlib, seaborn

Aestroid Risk Classification Notebook

- Developed a machine learning model to classify asteroids as **hazardous or non-hazardous** based on orbital and physical parameters.
- Implemented and compared **Random Forest** and **XGBoost** models, achieving highest accuracy of **82%** using XGBoost.
- Improved minority class detection (hazardous asteroids), achieving **recall up to 48%**.

Tech stack- XG Boost, Matplotlib, Seaborn, Classification Models

Achievements and Leadership

First Position - Asteroid Risk Classification Challenge View
Winner of Intra-IIT Machine Learning Challenge. Developed classification model using advanced feature engineering, model selection, and hyperparameter optimization techniques.

Third Position - Summer ChainRxn Bootcamp View
Secured third position in web development bootcamp. Developed animated front-end landing page with modern UI/UX principles.

MATLAB Machine Learning Certifications Cert 1 — Cert 2
Completed MATLAB Machine Learning Onramp and Classification Models specialized courses.

Core Member - Blockchain Society (BIB) IIT Bhilai
Core member of Blockchain at IIT Bhilai technical society. Organized workshops and hackathons .